

Small modular reactors for data centers remain a conditional low-carbon strategy

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ARTICLE INFO

Keywords:
Small modular reactor
Absorption cooling
Data center
Techno-economic analysis
Mixed-integer linear programming
Thermal integration

ABSTRACT

Data centers are an increasingly large and continuous electricity load, and their growth is intensifying demand for firm, low-carbon supply. Colocated small modular reactors (SMRs) with absorption cooling have been proposed to meet that demand while reusing reactor heat, but their economics have rarely been tested with hourly cooling physics, multi-year electricity and gas prices, current tax-credit policy, and realistic unit-size matching. Here we co-optimize the annualized cost of a 200 MW_e hyperscale data center in ERCOT at hourly resolution. We compare grid supply, a BWRX-300 reactor, the same reactor with a double-effect LiBr absorption chiller driven by mid-pressure turbine extraction, and an on-site natural gas combined cycle. The 100 scenarios span 2022–2024 prices, reactor and chiller capital, carbon price, financing and campus size, with the Inflation Reduction Act Section 45U nuclear production tax credit applied throughout. With the credit and nth-of-a-kind reactor capital, both SMR configurations fall below grid cost, by 25 and 22% with no carbon price, opening a viable region that capital-only analyses miss. This result depends on reaching that capital trajectory. At mid-range capital the two SMR configurations still cost \$192 and \$195 M/yr against \$78 M/yr for grid procurement even after the credit. Among the on-site options, only the gas plant undercuts grid supply at these conditions, by 29%. Pricing carbon near \$150/tCO₂ is a weaker second route at mid-range capital. That value is extrapolated beyond the simulated range and depends on how grid exports are credited. Absorption cooling is close to cost-neutral, changing total cost by \$2.4 M/yr relative to an SMR-only plant. Colocated SMR with absorption cooling is therefore a conditional low-carbon strategy, competitive where nth-of-a-kind reactor capital, the production tax credit, adequate load and a favorable market coincide.

Nomenclature

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Sets and indices

$t \in \mathcal{T}$	hourly time step over an annual modeling horizon, $ \mathcal{T} = 8760$
$c \in \mathcal{C}$	system configuration, $\mathcal{C} = \{0, 1, 2, 3\}$

Decision variables (hour t)

$P_{rx}(t)$	reactor thermal output (MW_{th})
$P_{tg}(t)$	turbine gross electric output (MW_e)
$P_{tn}(t)$	turbine net electric output (MW_e)
$Q_{ext}(t)$	mid-pressure steam extraction to absorber (MW_{th})
$Q_a(t)$	absorption-chiller cooling delivered (MW_c)
$P_v(t)$	vapor-compression chiller (VCC) electric input (MW_e)
$Q_v(t)$	VCC cooling delivered (MW_c)
$P_g^+(t)$	grid import (MW_e)
$P_g^-(t)$	grid export (MW_e)
$P_{ng}(t)$	NGCC electric output, Case 3 (MW_e)
$B^+(t)$	BESS charging power (MW_e)
$B^-(t)$	BESS discharging power (MW_e)
$E_B(t)$	BESS state of charge (MWh)

Abbreviations

ATB	NREL Annual Technology Baseline
BWRX-300	GE–Hitachi 870 MW_{th} / 270 MW_e Boiling Water SMR
CAPEX	overnight capital expenditure
CRF	capital recovery factor
EPBT	energy payback time
ERCOT	Electric Reliability Council of Texas
FOAK / NOAK	first- / nth-of-a-kind reactor capital cost
FOM, VOM	fixed / variable operations & maintenance
M	grid-cost margin, $1 - \text{TAC}_c / \text{TAC}_0$ (positive = cheaper than grid)
LCOE	levelized cost of electricity ($\$/\text{MWh}_e$)
LiBr– H_2O	lithium-bromide / water absorption pair
LMP	locational marginal price
NGCC	natural gas combined cycle
PTC	production tax credit (IRA Section 45U)
PUE	power usage effectiveness
SMR	small modular reactor
TAC	total annualized cost ($\$/\text{yr}$)
VCC	vapor-compression chiller
WACC	weighted-average cost of capital

Parameters

$P_{IT}(t)$	data-center IT load (MW_e)
$\pi_g(t)$	ERCOT day-ahead LMP ($\$/\text{MWh}_e$)
$\theta(t)$	ERCOT hourly average carbon intensity ($\text{g CO}_2/\text{kWh}_e$)
$T_{wb}(t)$	Houston wet-bulb temperature ($^\circ\text{C}$)
η_r	turbine rated electric efficiency at zero extraction (–)
α_w	Willans extraction penalty ($\text{MW}_e/\text{MW}_{\text{th}}$)
ξ_{aux}	turbine auxiliary fraction (–)
PUE	power usage effectiveness (–)
η_{ch}	chilled-water chain efficiency (–)
$\text{COP}_a(T_{wb})$	absorption chiller COP (–)
$a(t)$	absorption availability (a_m operating, 0 crystallization)
a_m	absorption maintenance availability (–)
COP_v	VCC full-load coefficient of performance (–)
$f_v(\cdot)$	VCC part-load electricity map (MW_e)
β_a	absorber parasitic per cooling output (kW_e/kW_c)
η_{rt}	BESS round-trip efficiency (–)
ρ_B	BESS hourly self-discharge (–)
Cap_E	BESS energy capacity (MWh)
$\overline{\text{CF}}$	reactor annual capacity factor (–)
$\overline{P}_{rx}, \underline{P}_{rx}$	reactor max / min thermal output (MW_{th})
$\Delta P_{rx}^{\max}$	reactor ramp limit per hour (MW_{th})
$\overline{P}_{pcc}$	point-of-common-coupling capacity (MW_e)
CAPEX_k	overnight capital coefficient of asset k
FOM_k	fixed O&M coefficient of asset k
VOM_k	variable O&M coefficient of asset k
CRF_k	capital recovery factor of asset k (–)
N_k	engineering life of asset k (yr)
π_f	delivered nuclear fuel cost ($\$/\text{MWh}_{\text{th}}$)
$\pi_{ng}(t)$	delivered natural-gas price for NGCC (Henry Hub + basis; $\$/\text{MMBtu}$)
h_{ng}	NGCC full-load HHV heat rate ($\text{MMBtu}/\text{MWh}_e$)
η_{hhv}	NGCC full-load HHV net efficiency (–)
$\overline{P}_{ng}$	NGCC rated electric capacity (MW_e)
$\phi_{ng}(\ell)$	NGCC part-load heat-rate multiplier (–)
$\bar{c}, \pi_1, \pi_2$	45U full credit and phaseout breakpoints ($\$/\text{MWh}_e$)
π_{co_2}	carbon price ($\$/\text{tCO}_2$)
Δt	hourly time step length (h)
γ_{rx}	reactor LCA emissions ($\text{g CO}_2/\text{kWh}_e$)
γ_{ng}	NGCC lifecycle emissions ($\text{g CO}_2\text{-eq}/\text{kWh}_e$)
Θ	kg-to-tonne unit conversion ($= 10^{-3}$)

1. Introduction

U.S. data centers are reshaping electricity demand after a decade in which efficiency gains moderated global data-center energy growth [1]. Consumption at U.S. data centers nearly doubled between 2019 and 2023 and is projected by the Lawrence Berkeley National Laboratory to reach 6.7%–12% of total U.S. electricity by 2028 [2]. The International Energy Agency projects global data-center electricity consumption to roughly double to 945 TWh by 2030, with the United States and China accounting for nearly 80% of that growth [3, 4]. The Electric Reliability Council of Texas (ERCOT), which serves a substantial share of the new U.S. data-center pipeline, expects its large-load interconnection queue to exceed 138 GW by 2030, creating a need for firm, dispatchable capacity beyond what variable renewables can supply alone [5]. The 2026 U.S. Annual Energy Outlook places much of this growth in regions managed by ERCOT and PJM, where transmission queues and capacity adequacy already constrain interconnection timelines [6].

Data-center loads impose an unusual set of requirements on the supplying generator. The workload is highly continuous, with capacity factors close to one, and modern hyperscale facilities require power availability at five nines or higher. Cooling is the dominant non-IT energy use, typically accounting for 20%–40% of total facility consumption, and it increases sharply with ambient wet-bulb temperature in inland sites such as Texas [7, 8]. These characteristics make data centers a natural counterpart to firm, low-carbon generators with baseload operating patterns. Small modular reactors (SMRs), in particular, have re-entered procurement and planning discussions because they provide dispatchable carbon-free output at unit sizes between roughly 50 MW_e and 300 MW_e, comparable to a single hyperscale campus [9–12].

Two streams of prior work converge on the system studied here. The first examines whether SMRs can supply data-center electricity at a cost competitive with grid supply. Asuega et al. [12] performed a Monte Carlo techno-economic analysis of light-water, gas-cooled, and molten-salt SMR configurations against natural gas combined cycle and conventional large reactors. They found that a 12 × 77 MW_e light-water SMR plant has a lower mean overnight capital cost and a shorter construction timeline than a single 1 GW reactor but remains above grid prices in most market regions without a carbon constraint. Recent nuclear-data-center studies extend this question to mixed SMR and distributed-energy portfolios, optimized cooling, and integrated power-thermal architectures [13–16]. The Idaho National Laboratory meta-analysis of 48 advanced-reactor cost data sets reaches a similar conclusion, with capital cost estimates dispersed widely between first-of-a-kind and nth-of-a-kind cases [17]. Broader SMR finance reviews and nuclear construction-cost studies point to the same sensitivity to learning, standardization, project delivery, and financing structure [9, 18]. The U.S. Department of Energy Pathways to Commercial Liftoff: Advanced Nuclear report consolidates this view and identifies long-term power purchase agreements with hyperscale technology firms as a financing mechanism that may lower effective capital cost over time through learning-by-doing [10]. Several 2024 and 2025 industry agreements between hyperscalers and reactor developers, including the Microsoft-Constellation Three Mile Island restart and the Google-Kairos Power 500 MW deal, reflect a shift from policy-driven to commercially driven nuclear deployment for data-center supply [19]. The 2024 NREL Annual Technology Baseline distinguishes 300 MW_e SMR capital cost among first-of-a-kind, mid, and nth-of-a-kind trajectories, with the gap between the two extremes exceeding a factor of six in 2024 USD [11].

The second stream studies cooling-side thermal integration. Reviews of data-center cooling technology identify low-grade waste heat as an underused resource but also emphasize that recovery value depends on sink temperature, operating hours, and auxiliary power [20, 21]. Thermodynamic and thermo-economic analyses of absorption chillers driven by data-center waste heat show that technically feasible cooling savings can be eroded by heat-exchanger approach temperatures, chiller capital cost, and low utilization during mild hours [22, 23]. Cooling consumes 20% to 40% of data-center energy in air-cooled facilities, and single-effect lithium-bromide water (LiBr-H₂O) absorption chillers convert low-grade heat into chilled water at coefficients of performance (COP) near 0.7 [24]. Amiri et al. [24] reported annual energy savings of 4.3 GWh to 13.0 GWh for a 4.5 MW to 13.5 MW data center retrofitted with an absorption chiller driven by IT-side waste heat, with corresponding emissions savings of 3,068 tCO₂ to 9,208 tCO₂ per year. Cui et al. [25] proposed a switchable multistage absorption design targeting ultra-low-grade waste heat below conventional driving-temperature requirements. In the broader absorption-cooling literature, double-effect designs reach COP between 1.1 and 1.4 but require higher driving temperatures, typically above 130 °C, that are more easily obtained from a steam-cycle reactor than from server-rack exhaust [26, 27]. Across both streams, the absorption submodel is sensitive to ambient wet-bulb temperature: a higher cooling-water inlet temperature drives the LiBr solution toward the crystallization line, forcing a switch to vapor-compression backup [25, 27]. In Texas, this de-rating concentrates in July and August afternoons, when wet-bulb temperatures regularly exceed 26 °C.

Methodologically, the boundary-analysis framing here draws on co-location and transactive-energy studies of wind–hydrogen systems, Texas hydrogen pathways and facility–microgrid siting [28–30]. Related demand-side work using real-time greenhouse-gas intensity motivates the time-resolved carbon accounting adopted here [31]. None of these studies models a nuclear steam cycle coupled to data-center cooling.

Together these two streams leave the siting and procurement decision underdetermined. Studies of thermally integrated absorption cooling have largely treated heat reuse as a categorical benefit, comparing a no-recovery baseline against a fully integrated configuration without identifying the technology- and market-specific conditions under which integration actually pays. Hourly cooling physics, including wet-bulb-driven COP de-rating and crystallization downtime, has rarely been combined with multi-year market data spanning the recent range of Henry Hub gas prices, ERCOT locational marginal prices, reactor capital-cost trajectories and current production-tax-credit policy. And because sensitivity analyses in this setting are usually one-dimensional, they offer no multi-axis envelope that maps a candidate site or financing structure onto a competitive or non-competitive region. Without such an envelope, claims about the viability of SMR with absorption cooling rest heavily on the chosen baseline year and reactor capital-cost path.

Here we present a techno-economic boundary analysis of a colocated SMR with double-effect absorption cooling, driven by cascaded HP–LP turbine extraction, for a 200 MW_e hyperscale data center on the ERCOT grid. A mixed-integer linear program co-optimizes annualized total cost across power generation, cooling supply and battery storage at hourly resolution over 8760-hour annual horizons, using year-resolved ERCOT day-ahead prices, Henry Hub gas prices and Houston wet-bulb temperatures from 2022 to 2024, and applying the Inflation Reduction Act Section 45U nuclear production tax credit to the reactor cases. We compare four configurations: grid-only supply with vapor-compression chillers (Case 0); a BWRX-300 plant with vapor-compression chillers (Case 1); a BWRX-300 plant coupled to a double-effect LiBr–H₂O absorption chiller driven by mid-pressure turbine extraction, with vapor-compression backup and a crystallization gate (Case 2); and an on-site natural gas combined cycle plant with vapor-compression chillers (Case 3). A multi-axis sensitivity design exposes each configuration to ranges in data-center load, power usage effectiveness, market year, battery deployment, reactor capital-cost trajectory, absorption-chiller capital cost, carbon price and cost of capital.

The analysis makes three advances. It couples an hourly absorption submodel to multi-year ERCOT price, Henry Hub and reactor-capital trajectories within a single co-optimization, under current Section 45U tax-credit policy. The submodel resolves wet-bulb-driven COP de-rating and a LiBr crystallization gate. This co-optimization maps the cost-competitiveness envelope of SMR with absorption cooling jointly over reactor and absorption-chiller capital cost. It separates this jointly tested capital-cost frontier from one-axis movements toward parity along the policy, financing, market-year and unit-size axes. And it converts the resulting envelope into procurement-relevant boundary conditions that generalize beyond ERCOT to other markets, climates and data-center sizes. Scenario definitions and input provenance are documented in the Supplementary Information.

The remainder of this paper is organized as follows. Section 2 describes the system configurations, the optimization formulation, the absorption submodel, and the data inputs. Section 3 presents the results and discusses the boundary conditions for thermal integration and the implications for site selection and procurement. Section 4 concludes.

2. Methodology

2.1. System configurations

The analysis compares four configurations that supply electricity and chilled water to a single 200 MW_e rated hyperscale data center in Houston, Texas. Each configuration uses the same data-center load profile, vapor-compression chiller technology when required, and financial parameters; the configurations differ only in the electricity supply pathway and in whether the cooling chain uses recovered reactor heat. Figure 1 summarizes the cogeneration mechanism, the grid and storage interfaces, and the boundary conditions varied in the sensitivity analysis. Cases 0, 1 and 3 are reduced versions of this layout in which selected blocks are disabled. Table 1 summarizes the four configurations.

Case 0 represents the conventional grid-supplied data center. All electricity is purchased from ERCOT at the day-ahead locational marginal price, and all cooling is produced by a water-cooled centrifugal vapor-compression chiller with a Houston-derated coefficient of performance of 3.2. This case serves as the benchmark for the grid-cost margin reported throughout the paper. Case 1 replaces grid imports with on-site BWRX-300 generation while retaining the same VCC. Heat at the turbine condenser is rejected to the cooling tower and is not recovered. Bidirectional grid exchange is retained so that excess electricity can be sold to ERCOT during low-load hours and grid imports can cover

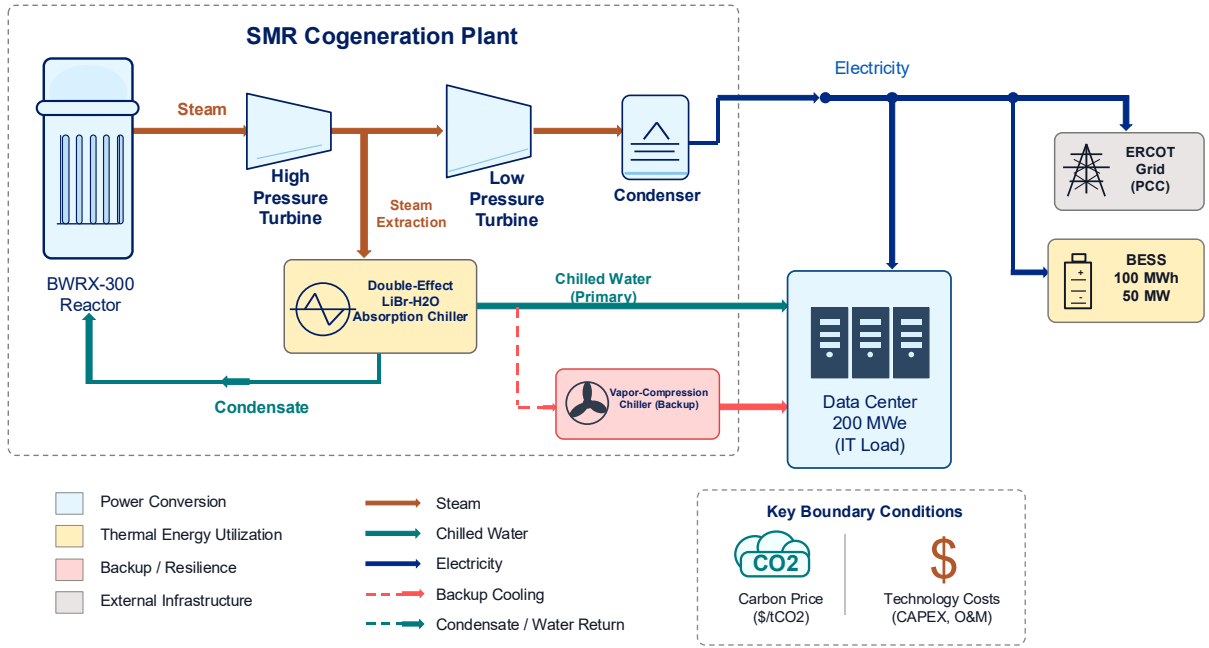


Figure 1: Conceptual mechanism for SMR-powered data-center cooling. Steam from the BWRX-300 reactor first expands through the turbine train; a mid-pressure extraction tap diverts heat to a double-effect LiBr–H₂O absorption chiller, while remaining electricity, grid exchange, battery storage, and vapor-compression backup serve the variable IT load of the 200 MWe rated campus (mean 94.7, peak 142.4 MWe). The lower inset highlights representative boundary conditions, including carbon price and technology costs; the full sensitivity set additionally varies cooling demand, weather year, storage and financing assumptions.

Table 1

System configurations modeled in this study. ✓denotes that the block is enabled; – that it is disabled. PCC = point of common coupling with ERCOT.

		Case 0	Case 1	Case 2	Case 3
Generation	BWRX-300 SMR + steam turbine	–	✓	✓	–
	NGCC on-site, F-class	–	–	–	✓
	ERCOT grid (PCC, 300 MW)	✓ (import only)	✓ (bi-directional)	✓ (bi-directional)	–
Cooling	VCC, water-cooled	✓ (primary)	✓ (primary)	✓ (backup)	✓ (primary)
	Double-effect LiBr absorber	–	–	✓	–
Storage	Li-ion BESS, 100 MWh / 50 MW	–	sensitivity	sensitivity	–

residual demand during planned outages. Case 2 adds a double-effect LiBr–H₂O absorption chiller fed by a mid-pressure extraction tap on the high-pressure (HP) turbine. The cascade keeps the low-pressure (LP) turbine in service, so diverted steam incurs a Willans penalty rather than the full turbine efficiency penalty. A 160 MWe vapor-compression backup chiller remains installed, the same capacity as the primary chiller in the VCC-only cases, because it must carry the full cooling load whenever the absorption unit shuts down at the LiBr crystallization threshold. Case 3 replaces the SMR with a 200 MWe on-site F-class natural gas combined cycle, retains the VCC, and disconnects from the grid to isolate the value of replacing grid procurement with a fossil-fueled on-site alternative.

The data-center IT load follows an annual hourly trace synthesized from a 200 MWe hyperscale reference profile with a peak hour of 142.4 MWe, an annual average of 94.7 MWe, and an annual utilization of 0.67. The same trace is used in every case and every scenario; the IT load is not a decision variable. We treat the full IT electrical draw as heat

that the cooling system must reject, so the hourly cooling demand is

$$Q_{cool}^{dem}(t) = \frac{P_{IT}(t)}{\eta_{ch}}, \quad (1)$$

where $\eta_{ch} = 0.90$ accounts for the chilled-water distribution chain. Each scenario is labeled by its full-load vapor-compression PUE, $1 + 1/(\eta_{ch} \text{COP}_v)$, which is 1.35 at the baseline COP of 3.2. The dispatch-realized annual PUE, $1 + P_{cool}/P_{IT}$, is slightly lower for the vapor-compression cases, near 1.33, because the part-load COP curve runs more efficiently than its full-load point. It falls to about 1.05 in Case 2, where steam-driven absorption supplies most of the cooling and leaves little electric cooling load. The cooling-efficiency sensitivity in Section 3.2 sweeps the chiller COP so that this full-load PUE label spans $\{1.10, 1.30, 1.50\}$, the range between high-efficiency and legacy hyperscale practice [7, 8].

2.2. Optimization formulation

The four cases are formulated as separate dispatch problems on the same 8760-hour horizon. Cases 0 and 3 reduce to closed-form annual cost evaluations because the absence of storage and heat recovery makes the hourly decisions independent. Cases 1 and 2 are mixed-integer linear programs: the only integer structure is the special-ordered-set (SOS2) representation of the vapor-compression chiller part-load curve described below, which is non-convex and cannot be relaxed; no binary commitment variables are introduced, because round-trip losses in the battery block already make simultaneous charge and discharge unprofitable. The same accounting structure is applied to all four cases so that cross-case comparisons remain on a common basis.

Reactor and turbine block (Cases 1 and 2). The BWRX-300 reactor is modeled as a continuous thermal source with bounded ramping and an annual capacity-factor constraint. The reactor block enforces

$$\underline{P}_{rx} \leq P_{rx}(t) \leq \overline{P}_{rx}, \quad \forall t \in \mathcal{T}, \quad (2)$$

$$\left| P_{rx}(t) - P_{rx}(t-1) \right| \leq \Delta P_{rx}^{\max}, \quad \forall t \geq 1, \quad (3)$$

$$\sum_{t \in \mathcal{T}} P_{rx}(t) \geq \overline{\text{CF}} \cdot \overline{P}_{rx} \cdot |\mathcal{T}|, \quad (4)$$

with $\overline{P}_{rx} = 870 \text{ MW}_{\text{th}}$, $\underline{P}_{rx} = 435 \text{ MW}_{\text{th}}$ (50% minimum-load fraction from the BWRX-300 design basis [32]), $\Delta P_{rx}^{\max} = 522 \text{ MW}_{\text{th}}/\text{h}$ corresponding to a 1 %/min ramp, and $\overline{\text{CF}} = 0.92$ from the mature U.S. light-water-reactor fleet average [33]. The annual capacity-factor constraint sets an annual energy floor equivalent to roughly one month of downtime, without modeling the timing of individual outages or introducing unit-commitment binaries.

The Rankine steam cycle is represented by a cascaded HP–LP turbine with a Willans-line extraction penalty. Letting $\eta_r = 0.345$ be the rated electric efficiency at zero extraction and $\alpha_w = 0.083 \text{ MW}_e/\text{MW}_{\text{th}}$ the marginal electricity lost per MW_{th} of mid-pressure steam diverted to the absorber:

$$P_{tg}(t) = \eta_r P_{rx}(t) - \alpha_w Q_{ext}(t), \quad (5)$$

$$P_{in}(t) = (1 - \xi_{aux}) P_{tg}(t), \quad (6)$$

with $\xi_{aux} = 0.10$. Because diverted steam still expands through the HP turbine before extraction, the Willans coefficient is roughly one order of magnitude smaller than charging the full turbine efficiency against extracted heat, as would be appropriate for a parallel main-steam tap.

Absorption-chiller submodel (Case 2). The double-effect LiBr–H₂O absorption chiller is described by a time-varying coefficient of performance and a binary availability indicator that captures LiBr crystallization risk [26, 27]. With $T_{wb,0} = 26^\circ\text{C}$ as the Houston design wet-bulb, $\text{COP}_a^0 = 1.10$ as the COP at the design wet-bulb, $\delta = 0.015^\circ\text{C}^{-1}$ as the linear de-rating slope, and $\overline{\text{COP}}_a = 1.30$ as the nameplate cap, the chiller COP at wet-bulb $T_{wb}(t)$ is

$$\text{COP}_a(T_{wb}(t)) = \max \left\{ \min \left[\text{COP}_a^0 - \delta (T_{wb}(t) - T_{wb,0}), \overline{\text{COP}}_a \right], 0.5 \right\}. \quad (7)$$

The lower bound of 0.5 is a numerical safeguard; Houston wet-bulbs in the weather record do not reduce a physical chiller to this value because the crystallization gate engages first. With $\Delta T_{cw} = 5$ K as the cooling-tower approach, $T_{cry} = 32$ °C as the LiBr-crystallization-line cooling-water inlet, and $a_m = 0.95$ a maintenance availability that derates the unit to account for scheduled servicing, the availability indicator is

$$a(t) = \begin{cases} a_m & \text{if } T_{wb}(t) + \Delta T_{cw} \leq T_{cry}, \\ 0 & \text{otherwise,} \end{cases} \quad (8)$$

so the absorption chiller is forced fully offline whenever the cooling-water inlet would exceed the LiBr crystallization line, and otherwise operates at its maintenance-derated availability. Equivalently, with the $\Delta T_{cw} = 5$ K tower approach this is a wet-bulb gate at $T_{cry} - \Delta T_{cw} = 32 - 5 = 27$ °C, the value marked in Figure 2a. Delivered absorption cooling is

$$Q_a(t) = \text{COP}_a(T_{wb}(t)) \cdot a(t) \cdot Q_{ext}(t), \quad (9)$$

167 and a parasitic electric load $\beta_a Q_a(t)$ with $\beta_a = 0.020$ kW_e/kW_c powers solution and refrigerant pumps. The vapor-
168 compression backup carries the residual cooling load whenever $a(t) = 0$.

Cooling, BESS and grid blocks. Vapor-compression cooling links its electric input to its cooling output through a part-load coefficient of performance. A water-cooled centrifugal chiller gains efficiency at reduced load, with the coefficient of performance peaking near 40–50% of rated duty before declining at very low load, following the integrated part-load value of ANSI/AHRI Standard 550/590 [34]. The chiller electricity is therefore a non-convex, piecewise-linear function of the cooling delivered, $P_v(t) = f_v(Q_v(t))$, anchored at the Houston-derated full-load value $\text{COP}_v = 3.2$. In the closed-form Cases 0 and 3 the chiller is evaluated directly at the realized load fraction; in the nuclear Cases 1 and 2 the same curve enters the optimization as an SOS2 piecewise map, the single source of integer structure in the model. The lithium-ion BESS, sized at 100 MWh / 50 MW (C/2), follows the standard state-of-charge dynamics

$$E_B(t) = (1 - \rho_B) E_B(t-1) + (\sqrt{\eta_{rt}} B^+(t) - B^-(t)/\sqrt{\eta_{rt}}) \Delta t, \quad (10)$$

169 with cyclical closure $E_B(|\mathcal{T}|) = E_B(0)$, $\eta_{rt} = 0.85$, $\rho_B = 4.17 \times 10^{-5} \text{ h}^{-1}$, and SOC bounds $[0.05, 0.95] \text{ Cap}_E$. Grid
170 exchange is bounded by the point of common coupling, $0 \leq P_g^+(t)$, $P_g^-(t) \leq \bar{P}_{pcc}$, with $\bar{P}_{pcc} = 300$ MW for the baseline
171 campus; in the size-matching sweep the interconnect scales with the data-center load, since it is part of the data-center
172 build rather than the fixed reactor.

Energy balances. Two equality constraints hold every hour. The electric balance equates supply, BESS discharge and grid imports to the data-center IT load, chiller drives, absorber parasitics, BESS charge and grid exports:

$$\begin{aligned} P_{in}(t) + P_g^+(t) + B^-(t) &= P_{IT}(t) + P_v(t) + \beta_a Q_a(t) \\ &+ P_g^-(t) + B^+(t). \end{aligned} \quad (11)$$

The cooling balance equates total cooling supplied to the cooling demand of Eq. (1):

$$Q_a(t) + Q_v(t) = Q_{cool}^{dem}(t). \quad (12)$$

173 For Cases 0, 1 and 3 the absorption variables $Q_a(t)$, $Q_{ext}(t)$ are fixed to zero; for Case 3 the reactor and turbine variables
174 are removed and the IT and chiller load is served entirely by the NGCC block at a full-load HHV efficiency $\eta_{hhv} = 0.495$,
175 with the part-load heat-rate penalty of Eq. (18).

Objective. The optimization minimizes total annualized cost (TAC) summed over seven accounting buckets:

$$\begin{aligned} \min \text{ TAC} &= \sum_k \text{CRF}_k \text{CAPEX}_k S_k + \text{FOM} + \text{VOM} \\ &+ \Phi_{fuel} + \Phi_{grid} + \Phi_{co2} - \Phi_{ptc}, \end{aligned} \quad (13)$$

where the first term annualizes overnight capital across reactor, turbine, absorption and VCC chillers, BESS and NGCC sized at S_k , each amortized over its own engineering life N_k through an asset-specific capital recovery factor CRF_k

(reactor and turbine 20 yr, chillers 25 yr, NGCC 30 yr, battery 15 yr); $FOM = \sum_k FOM_k S_k$ aggregates the fixed O&M, and v_x denotes the variable-O&M coefficient of stream x (Table 3); the variable terms are

$$\begin{aligned} VOM = & \sum_t [v_{rx} P_{in}(t) + v_v Q_v(t) + v_a Q_a(t)] \Delta t \\ & + \sum_t v_B (B^+(t) + B^-(t)) \Delta t, \end{aligned} \quad (14)$$

$$\Phi_{fuel} = \pi_f \sum_t P_{rx}(t) \Delta t, \quad (15)$$

$$\Phi_{grid} = \sum_t \pi_g(t) [P_g^+(t) - P_g^-(t)] \Delta t, \quad (16)$$

$$\Phi_{co_2} = \pi_{co_2} \Theta \sum_t [\gamma_{rx} P_{in}(t) + \theta(t) (P_g^+(t) - P_g^-(t))] \Delta t. \quad (17)$$

For Case 3, the same accounting buckets are used but the reactor terms are replaced by the NGCC terms. The combined-cycle heat rate rises below the design point, so fuel use and combustion emissions are both scaled by a part-load multiplier $\phi_{ng}(\ell) \geq 1$ of the electrical load fraction $\ell(t) = P_{ng}(t)/\bar{P}_{ng}$, taken from a representative F-class combined-cycle characteristic and equal to one at full load. The VOM term uses $v_{ng} P_{ng}(t) + v_v Q_v(t)$, where $P_{ng}(t)$ is the NGCC electric output required by the IT and VCC loads, and the fuel and carbon terms become

$$\Phi_{fuel}^{ng} = \sum_t \pi_{ng}(t) h_{ng} \phi_{ng}(\ell(t)) P_{ng}(t) \Delta t, \quad (18)$$

$$\Phi_{co_2}^{ng} = \pi_{co_2} \Theta \sum_t \gamma_{ng} \phi_{ng}(\ell(t)) P_{ng}(t) \Delta t, \quad (19)$$

with full-load heat rate $h_{ng} = 3.412/\eta_{hhv}$ MMBtu/MWh_e and full-load lifecycle factor $\gamma_{ng} = 420$ g CO₂-eq/kWh_e, equal to 360 g/kWh_e direct combustion emissions plus 60 g/kWh_e upstream methane CO₂-equivalent [35]. Applying the same ϕ_{ng} to both terms holds the carbon content per unit of fuel constant. The carbon term Φ_{co_2} multiplies the case carbon footprint by a uniform price π_{co_2} and a unit-conversion factor $\Theta = 10^{-3}$ tCO₂/kg. Exports credit the case at the ERCOT hourly average carbon intensity $\theta(t)$, derived from the EIA-930 generation mix [36], to reflect the time-varying emissions of the grid power exchanged, following time-resolved hourly emissions accounting rather than a static annual-average factor [31]; this hourly average emission factor is used as a tractable proxy for the true marginal emission factor [37, 38]. Each asset is annualized over its own engineering life through $CRF_k = i(1+i)^{N_k}/[(1+i)^{N_k} - 1]$, with asset life N_k as listed above and baseline WACC $i = 6.7\%$; the battery additionally carries a 5% end-of-life salvage credit. The WACC sensitivity in Section 3.2 sweeps i over {5%, 6.7%, 10%}.

Nuclear production tax credit. The nuclear cases earn the zero-emission nuclear production tax credit (Section 45U; 26 U.S.C. § 45U), enacted in the Inflation Reduction Act of 2022 [39], on their generation [40, 41]. At the prevailing-wage rate the credit is \$15/MWh_e (1.5¢/kWh_e) and is subject to the statutory gross-receipts phaseout: it holds at the full value while the hourly market price π_g stays at or below the threshold $\pi_1 = \$25/\text{MWh}_e$ (2.5¢/kWh_e), then declines by \$0.80 for each \$1/MWh_e above it, reaching zero at $\pi_2 = \$43.75/\text{MWh}_e$ (4.375¢/kWh_e),

$$c_{45U}(\pi) = \begin{cases} \bar{c}, & \pi \leq \pi_1, \\ \bar{c} \left(1 - \frac{\pi - \pi_1}{\pi_2 - \pi_1}\right), & \pi_1 < \pi < \pi_2, \\ 0, & \pi \geq \pi_2, \end{cases} \quad (20)$$

with full credit $\bar{c} = \$15/\text{MWh}_e$. The annual credit is applied to net nuclear generation,

$$\Phi_{ptc} = \sum_t c_{45U}(\pi_g(t)) P_{in}(t) \Delta t, \quad (21)$$

using the hourly ERCOT price as the per-kWh gross-receipts proxy. The slope and breakpoints reproduce the statutory 16% reduction at the prevailing-wage multiplier. We treat the credit as a baseline policy condition, assuming a power-purchase-agreement structure in which the data center is an unrelated offtaker so that delivered nuclear generation

qualifies; statutory amounts are inflation-indexed after 2024 and the program runs through tax year 2032, both of which we hold at their nominal in-window values. Cases 0 and 3 carry no nuclear generation and therefore no credit.

The cross-case figure of merit used throughout the rest of the paper is the *grid-cost margin*

$$M_c = 1 - \frac{TAC_c}{TAC_0} \Big|_{\text{matched scenario}}, \quad (22)$$

defined as one minus the ratio of the case TAC to the Case 0 (grid-only) TAC at the same year, PUE and carbon price. A positive margin means the configuration is cheaper than grid procurement and a negative margin means it is more expensive; the magnitude is the relative cost difference. We report it for all four configurations as a common measure of cost competitiveness against the grid.

2.3. Data inputs

Inputs to the optimization fall into three families: *exogenous time series* (hourly traces of IT load, electricity price, average carbon intensity, and wet-bulb temperature), *equipment parameters* (capital and operating cost coefficients plus technical bounds for each plant block), and *financial parameters* (project lifetime and weighted-average cost of capital). All runs use 8760-hour annual operating horizons; leap-year raw files are trimmed to 8760 hourly records for cross-year comparability. The four characteristic input traces are shown in Figure 2; aggregated values and source citations are listed in Tables 2 and 3.

The Houston wet-bulb temperature distribution (Figure 2a) drives the absorption-chiller availability constraint in Equation (8). Monthly box plots from the Open-Meteo historical-weather archive [42] for the 2023 calendar year show that wet-bulb exceeds 26 °C (the absorption design anchor $T_{wb,0}$) in about 32% of summer hours and crosses the LiBr crystallization gate at 27 °C in about 5% of summer hours, concentrated in July and August afternoons. The ERCOT day-ahead LMP duration curves (Figure 2b), retrieved through gridstatus.io [43], capture the recent range of grid-procurement cost: 2022 represents the high natural-gas price regime, with an annual mean LMP of \$70/MWh_e and frequent excursions above \$200/MWh_e; 2023 was a moderate year at \$57/MWh_e; and 2024 declined to \$28/MWh_e as gas prices receded and renewable generation added supply. The IT-load profile (Figure 2c) follows a typical hyperscale shape with mild diurnal seasonality and an annual capacity factor of 0.67; peak load reaches 142.4 MW_e and annual mean load is 94.7 MW_e. The absorption-chiller COP_a(T_{wb}) curve (Figure 2d) parameterizes Equation (7): COP decreases linearly from the nameplate cap of 1.30 at cool ambient wet-bulb to 1.10 at the design anchor of 26 °C, after which the crystallization gate forces a binary cutoff and the VCC backup carries the cooling load.

Scalar inputs are summarized in Table 2, and the per-asset cost and technical parameters carried through the CAPEX, FOM, and VOM terms in Equation (13) are listed in Table 3. Reactor capital cost trajectories follow the NREL 2024 Annual Technology Baseline [11] at \$14,700/kW_e FOAK, \$7,615/kW_e Moderate, and \$2,250/kW_e NOAK. The absorption-chiller capital cost grid is anchored at \$750/kW_c baseline and brackets the literature range from \$450/kW_c (Bare-Low, Chinese OEM ex-works) to \$1,200/kW_c (Turnkey-High, Western EPC). The NGCC cost block follows the NREL ATB 2024 F-class Moderate values; natural-gas fuel cost is the hourly Henry Hub spot price [44] plus the Houston Ship Channel basis (mean \$0.10/MMBtu). The data audit trail and provenance summary are provided in the Supplementary Information.

2.4. Scenario design

Eight sensitivity groups are evaluated relative to the baseline 2023 / full-load PUE 1.35 / ATB-Mid CAPEX / WACC 6.7% configuration to probe the boundary conditions of the cogeneration economics. Table 4 lists the full run ledger, consisting of four baseline case runs plus 96 sensitivity runs.

The PUE sweep tests the sensitivity of the absorption value proposition to the cooling-load fraction. The year sweep covers the recent range of Henry Hub and ERCOT prices. The BESS sweep tests whether 100 MWh of storage adds value at current Li-ion capital costs in the nuclear LP cases, with Case 0 and Case 3 retained as no-storage control rows. The SMR CAPEX sweep covers the published FOAK-to-NOAK range. The two-dimensional SMR × absorption CAPEX grid (group G5) is the central scenario: it asks whether any joint capital-cost trajectory makes Case 2 cheaper than Case 0 at the 2023 baseline. The carbon-price sweep identifies the carbon price at which the nuclear cases cross the grid-only baseline. The WACC sweep isolates the financing effect on Case 2 with the operating parameters held constant. The data-center size sweep (group G8) scales the IT-load trace and data-center-side cooling / NGCC capacities from 0.5× to 3.0× while holding the commercial BWRX-300 unit fixed, isolating the effect of unit-size matching.

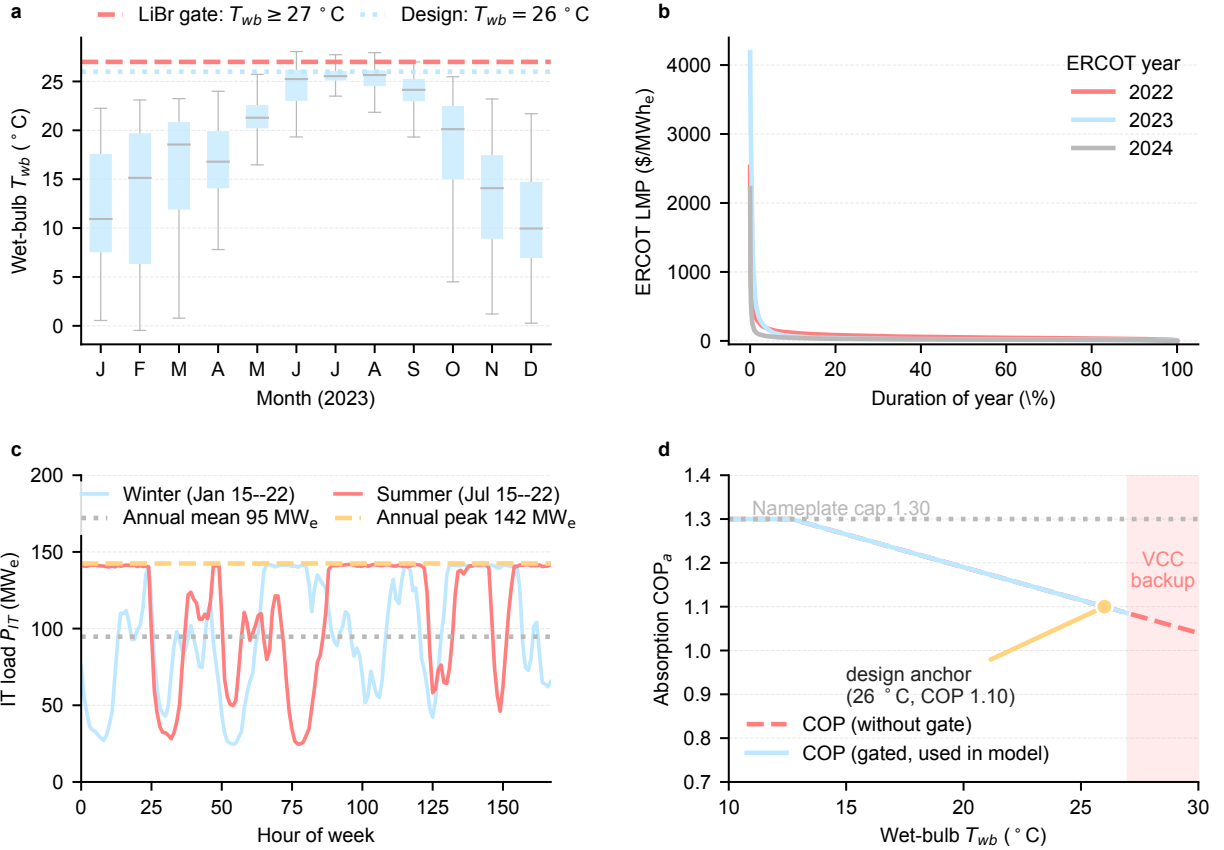


Figure 2: Hourly input traces driving the optimization. (a) Houston monthly wet-bulb distribution from the Open-Meteo historical-weather archive (year 2023). The dashed horizontal line marks the LiBr crystallization gate at $T_{wb} \geq 27^{\circ}\text{C}$; the dotted line marks the COP design anchor $T_{wb,0} = 26^{\circ}\text{C}$. (b) Duration curves of ERCOT day-ahead LMP for the three modeled years (2022 high-gas-price regime; 2023 moderate regime; 2024 low-LMP regime). (c) Data-center IT-load profile for a typical winter and summer week, with annual mean and peak overlaid. (d) Double-effect LiBr-H₂O absorption-chiller COP as a function of wet-bulb temperature; the dashed segment shows the linear de-rating below the crystallization gate, and the shaded band marks the backup-cooling region in which the VCC carries the cooling load.

3. Results and Discussion

Two facts organize the results. First, at 2023 mid-range reactor capital the SMR configurations cost roughly 2.5 times grid supply even after the production tax credit. Second, this gap is conditional rather than structural. At nth-of-a-kind reactor capital the same credit brings both configurations below grid cost with no carbon price. This opens a viable region along the lowest-capital edge of the joint capital-cost frontier. The baseline economics below establish the size of the gap and its operational sources, and the boundary sensitivities then identify the reactor-capital, market and policy conditions under which it closes.

3.1. Baseline economics and operational anatomy

Figure 3 decomposes the annualized cost of the four configurations at the 2023 ATB-Mid baseline, with the Section 45U credit applied to the nuclear cases. The grid-supplied data center (Case 0) carries a TAC of \$78.1 M/yr, of which 93% is the ERCOT electricity bill. The two SMR configurations remain far more expensive, at \$192.2 M/yr (Case 1) and \$194.6 M/yr (Case 2), even after the production tax credit returns \$26.4 and \$25.8 M/yr. Reactor capital recovery dominates these totals, and surplus-export revenue together with the credit offsets a little under half of the capital line. The on-site NGCC plant (Case 3) is the only configuration that beats grid supply at this baseline, at \$55.7 M/yr, a grid-cost margin of +29%, because the 2023 delivered gas price of \$2.63/MMBtu (Henry Hub plus the

Table 2

Input data sources, baseline values and units.

Quantity	Baseline value or range	Resolution	Source
ERCOT day-ahead LMP, Houston Hub	yearly mean \$28–70/MWh _e (2022–2024)	hourly	gridstatus.io
ERCOT average carbon intensity, ERCOT system	320–351 g CO ₂ /kWh _e annual mean (hourly AEF)	hourly	EIA-930 + UNECE 2022
Delivered natural gas (Henry Hub + basis)	\$2.35–6.53/MMBtu (annual mean)	hourly	EIA Henry Hub spot
Houston wet-bulb temperature	–0.5 to 28 °C annual range	hourly	Open-Meteo historical
Data-center IT load	142.4 MW _e peak / 94.7 MW _e mean	hourly	LBNL hyperscale benchmark
BWRX-300 capital cost	\$7,615/kW _e Moderate; \$14,700/kW _e FOAK; \$2,250/kW _e NOAK	scalar	NREL ATB 2024 [11]
BWRX-300 fixed O&M	\$121/kW _e /yr	scalar	NREL ATB 2024
BWRX-300 capacity factor	0.92 (fleet average)	scalar	EIA EPM Table 6.07.B [33]
BWRX-300 LCA emissions	12 g CO ₂ -eq/kWh _e	scalar	UNECE 2022
Absorption chiller capital cost	\$450–1,200/kW _e (\$750 baseline)	scalar	ASHRAE TC8.3 + vendor literature
VCC capital cost	\$250/kW _e	scalar	ASHRAE typical centrifugal
Li-ion BESS capital cost	\$405/kWh + \$1,620/kW	scalar	NREL ATB 2024
NGCC capital cost	\$1,330/kW _e	scalar	NREL ATB 2024 F-class Moderate
NGCC direct CO ₂	360 g CO ₂ /kWh _e + 60 g upstream CH ₄ (CO ₂ -eq)	scalar	EIA + Alvarez et al. (2018)
WACC	6.7% baseline; {5, 6.7, 10}% in scenario group G7	scalar	MIT ANP-201 nuclear-finance survey
Asset amortization life	20 yr reactor/turbine; 25 yr chillers; 30 yr NGCC; 15 yr battery	scalar	NREL ATB 2024
Nuclear PTC (Section 45U)	\$15/MWh _e full credit; phaseout \$25–43.75/MWh _e	scalar	26 U.S.C. 45U; IRS Form 7213

Houston Ship Channel basis) held the on-site fuel cost below the grid-procurement bill. The day-ahead price still fell below the NGCC running cost of about \$21/MWh_e in roughly 42% of hours (Supplementary Information), but the islanded Case 3 cannot import in those hours. Adding absorption cooling to the nuclear plant changes total cost by only \$2.4 M/yr (Case 2 minus Case 1). The absorption capital still slightly outweighs the cooling electricity it displaces, so the chiller is close to cost-neutral and does not lower data-center cost.

Sizing and the overbuild component of the margin. The margin at this baseline reflects the smallest commercial BWRX-300 unit (270 MW_e net at a 0.92 capacity factor), about 2.9 times the data center's 94.7 MW_e average IT load, so the reactor exports a structural surplus near 1.43 TWh/yr to ERCOT. The size-matching sweep (group G8; Figure 4) holds the reactor fixed while scaling the data-center load, cooling and grid interconnect from 0.5× to 3.0×. Scaling the campus to 2.5×, or 237 MW_e mean demand, brings Case 2 close to import-export balance (about +0.04 TWh/yr), yet it remains 62% more expensive than the matching grid-only baseline; at 3.0× the site becomes a net importer (about –0.42 TWh/yr) and the margin reaches –52%. The annual dollar gap to grid supply barely moves, from \$117 M/yr at 1.0× to \$121 M/yr at 3.0×, because the larger campus also raises the grid-only counterfactual. What improves is utilization intensity: Case 2 cost per MWh of IT load falls from \$235 at 1.0× to \$152 at 2.5× and \$143 at 3.0×, and the excess over grid supply falls from \$140 to \$58 and \$49/MWh_{IT}. Unit-size indivisibility therefore inflates the small-campus result, but eliminating overbuild still leaves the ATB-Mid reactor-capital penalty intact.

The same ranking persists across the three ERCOT years considered (Figure 5). In the cost–carbon plane, the nuclear cases occupy the low-carbon region (lifecycle 12 g CO₂/kWh_e from the UNECE 2022 reactor LCA [45], net of

Table 3

Detailed equipment cost and performance parameters used in the optimization. CAPEX is the overnight capital coefficient; FOM the fixed O&M coefficient; VOM the variable O&M coefficient. Each asset's CAPEX is annualized over its own listed lifetime through an asset-specific capital recovery factor; the battery additionally carries a 5% end-of-life salvage credit.

Asset	CAPEX	FOM (per yr)	VOM	Lifetime	Other technical parameters
BWRX-300 reactor (Cases 1, 2)	\$7,615/kW _e	\$121/kW _e	\$3.50/MWh _e	20 yr	870 MW _{th} / 270 MW _e net; $\eta_r = 0.345$; min-load fraction 0.50; ramp 1%/min; CF 0.92; fuel \$6/MWh _{th} ; LCA 12 g CO ₂ /kWh _e
Cascaded steam turbine (Cases 1, 2)	bundled into reactor OCC	bundled	bundled	20 yr	$\eta_r = 0.345$ rated at zero extraction; aux 10%; Willans slope $\alpha_w = 0.083$ MW _e /MW _{th}
Double-effect absorption chiller (Case 2)	\$450–1,200/kW _c (\$750 baseline)	\$20/kW _c	\$0.40/MWh _c	25 yr	nameplate cap COP 1.30; design $T_{wb,0} = 26^\circ\text{C}$, COP 1.10; de-rating 0.015/°C; crystallization $T_{cry} = 32^\circ\text{C}$ cooling-water inlet; tower approach 5 K; parasitic $\beta_a = 0.020$ kW _e /kW _c ; 160 MW _c nameplate
Vapor-compression chiller (Cases 0, 1, 2, 3)	\$250/kW _c	\$8/kW _c	\$0.50/MWh _c	25 yr	water-cooled centrifugal; nameplate COP 3.5; Houston-derated COP 3.2; 160 MW _c in all cases
Li-ion BESS (G3)	\$405/kWh + \$1,620/kW	\$25/kW	\$0.50/MWh	15 yr	100 MWh / 50 MW (C/2); round-trip $\eta_{rt} = 0.85$; self-discharge 0.1%/day; SOC bounds [0.05, 0.95]
NGCC (Case 3)	\$1,330/kW _e	\$29.3/kW _e	\$2.65/MWh _e	30 yr	200 MW _e F-class; HHV $\eta_{hhv} = 0.495$; direct CO ₂ 360 g/kWh _e ; upstream CH ₄ 60 g/kWh _e (CO ₂ -eq); Henry Hub + Houston Ship Channel basis \$0.10/MMBtu
ERCOT interconnection (Cases 0, 1, 2)	not capitalized	N/A	N/A	N/A	300 MW PCC bidirectional; flat substation interconnection in baseline

grid-export credit at the ERCOT hourly average carbon intensity), while the grid-only baseline anchors the low-cost region in the 2023 and 2024 years. Case 3 NGCC trades higher carbon emissions for lower cost; its 2023 point lies closest to the preferred bottom-left direction. The nuclear carbon-accounting panel shows that grid-export credits more than offset grid-import and reactor-lifecycle emissions in Cases 1 and 2, but none of the nuclear configurations reaches the low-cost portion of the Pareto frontier at zero carbon price.

Operational anatomy of the cogeneration case.

The hourly dispatch of Case 2 shows how the absorption chiller integrates with the reactor under winter and summer operating conditions (Figure 6a,b). In the winter week, the BWRX-300 runs near nameplate thermal output, consistent with its 92% annual capacity-factor constraint. The HP–LP turbine produces close to its full ~ 270 MW_e net output through most of the window, turning down to about 125 MW_e during the low-price hours. Chilled-water demand tracks the 25–142 MW_e winter IT load through $Q_{cool}^{dem} = P_{IT}/\eta_{ch}$, and the extraction tap supplies most of it as steam, with the vapor-compression backup contributing only on the highest-demand hours. In the July week, wet-bulb temperature climbs toward the 26 °C design anchor on the warm afternoons, derating the absorption COP toward its 1.10 design value. The week stays below the 27 °C crystallization gate (32 °C cooling-water inlet). The higher summer cooling

Table 4

Baseline and sensitivity scenario run ledger. Cells marked – indicate that the case is not included in that scenario group.

Group	Sweep	Levels	Runs	C0	C1	C2	C3
G0	Baseline	2023, full-load PUE 1.35, ATB-Mid CAPEX, WACC 6.7%	4	✓	✓	✓	✓
G1	Effective PUE	1.10, 1.30, 1.50	6	–	✓	✓	–
G2	ERCOT year	2022, 2023, 2024	12	✓	✓	✓	✓
G3	BESS	on, off	8	ctrl.	✓	✓	ctrl.
G4	SMR CAPEX	FOAK, ATB-Mid, NOAK	6	–	✓	✓	–
G5	SMR × Abs. CAPEX	5 × 5 grid	25	–	–	✓	–
G6	Carbon price	0, 50, 100 \$/tCO ₂	12	✓	✓	✓	✓
G7	WACC	5, 6.7, 10%	3	–	–	✓	–
G8	DC size	0.5–3.0× IT load	24	✓	✓	✓	✓
Total			100				

Note: In G3, “ctrl.” denotes duplicated control rows retained in the run ledger; BESS dispatch is optimized only for Cases 1 and 2 because Cases 0 and 3 are closed-form annual cost evaluations.

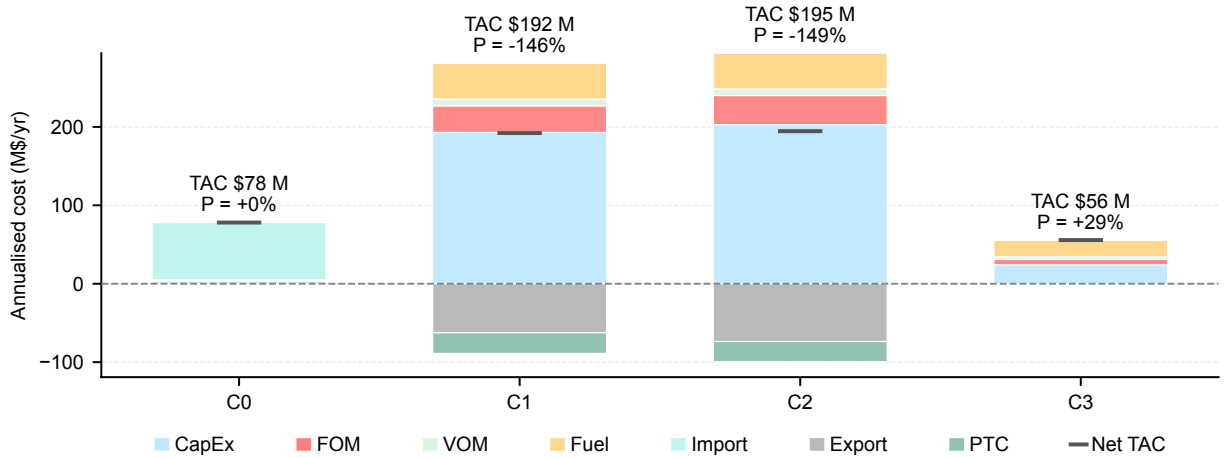


Figure 3: Total annualized cost decomposition for the four configurations at the 2023 ATB-Mid baseline (full-load PUE 1.35, WACC 6.7%). Each bar stacks capital recovery, fixed and variable O&M, fuel and grid imports; bars dipping below zero are grid-export revenues and, for the nuclear cases, the Section 45U production tax credit. The horizontal mark on each bar is the net TAC. Grid-cost margin is the relative cost difference from Case 0.

load nonetheless exceeds what the derated absorber can supply from steam, so the vapor-compression backup tops up to about 30 MW_c on the warmest afternoons. Across the full year the absorption chiller delivers 862 GWh_c and the vapor-compression backup 60 GWh_c, the latter spread over roughly 2870 hours of partial top-up that also cover the 5% maintenance derating of the absorber (Supplementary Information). The crystallization gate itself is rare: the availability indicator in Equation 8 forces the absorber fully offline for about 115 hours, concentrated in late June through August (annual peak wet-bulb 28 °C), during which the backup carries the entire cooling load of those hours (13.6 GWh_c).

The waterfall in Figure 6c decomposes the \$2.4 M/yr gap between Case 2 and Case 1 into the marginal components of adding absorption cooling to the nuclear-only plant. The chiller displaces \$18.2 M/yr of gross vapor-compression electricity at the ERCOT 2023 hourly import price. Against this credit stand the absorption capital plus fixed O&M of \$13.2 M/yr, the Willans-line electricity opportunity cost of \$4.1 M/yr from extracted steam, the VCC backup and top-up of \$2.3 M/yr, and a \$0.6 M/yr reduction in the production tax credit, since diverting steam to the absorber lowers the creditable turbine output; the monetized direct-site-water increment is only \$0.04 M/yr. A small –\$0.3 M/yr residual

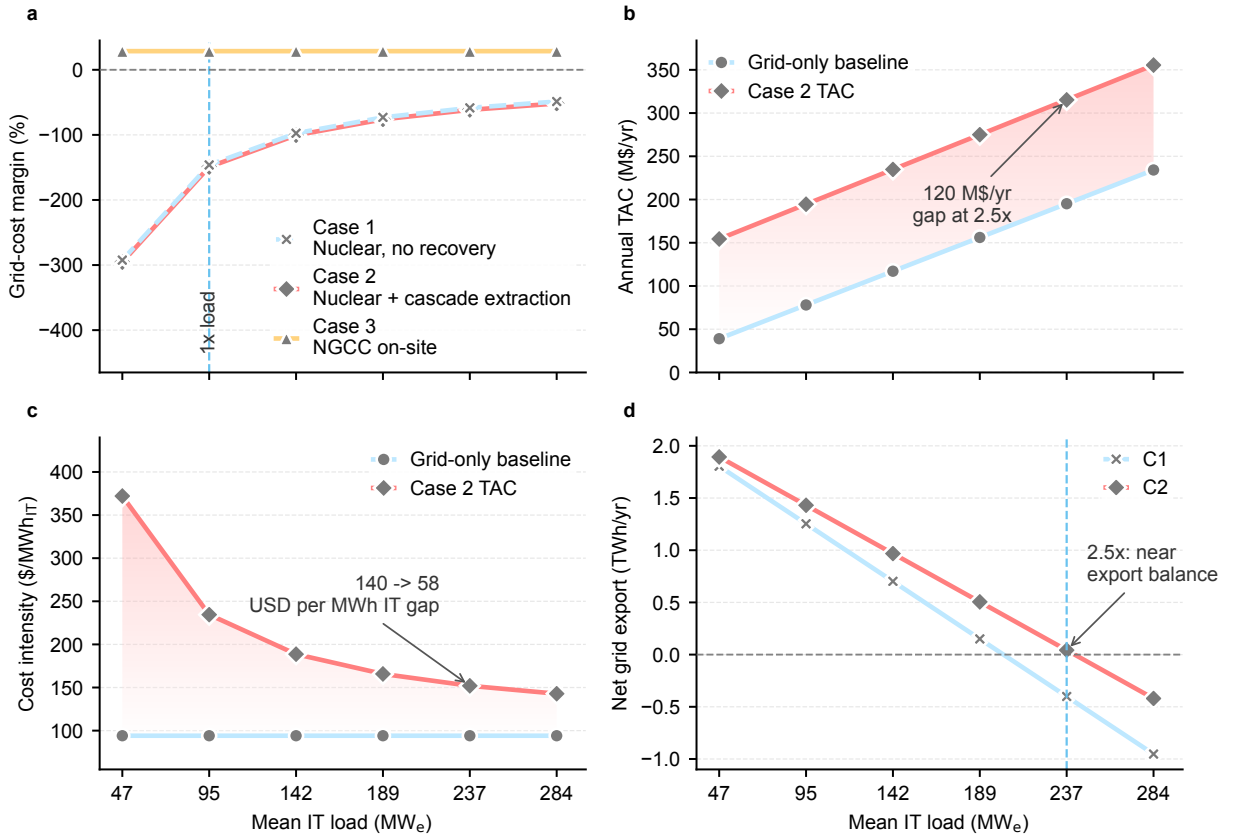


Figure 4: Data-center size matching sensitivity (G8). The IT-load trace and data-center-side cooling / NGCC capacities are scaled from 0.5× to 3.0× while the BWRX-300 reactor remains fixed at 270 MW_e net. (a) TAC-normalized grid-cost margin for the on-site cases against the matching grid-only baseline. (b) Case 2 TAC against the matching grid-only TAC, with the shaded area indicating the annual dollar gap. (c) Cost intensity per MWh of IT load; the shaded area is the excess Case 2 cost over grid supply. (d) Net grid export for the two nuclear cases; positive values indicate merchant export and negative values indicate net grid import. Arrow labels highlight the 2.5× load reference discussed in the text.

reconciles the component sum with the measured TAC difference. Credit and cost nearly cancel, so the chiller is close to break-even, and the residual penalty is a capital-cost problem rather than one of operating physics or water.

3.2. Boundary-condition sensitivities

The sensitivities below translate the baseline cost penalty into a boundary map. Figure 7 first isolates the cooling-side demand response and the marginal value of the absorption chiller. Figure 8 then collects the non-policy competitiveness boundaries in a single atlas: market year, reactor capital cost, financing and the joint reactor-absorber capital-cost frontier. This ordering separates one-axis movements toward parity from the two-axis capital-cost test that a procurement decision would require. Figure 9 then asks whether a carbon price can move the nuclear configurations across the grid-only baseline and ranks the remaining one-variable sensitivities.

Cooling-system efficiency (full-load PUE).

With the heat load fixed at P_{IT}/η_{ch} , the cooling-side lever is the efficiency of the electric chiller rather than the size of the duty. Figure 7 sweeps the chiller coefficient of performance so that the reported PUE spans {1.10, 1.30, 1.50} for Cases 1 and 2, with all other parameters at the 2023 ATB-Mid baseline, and places the same Case 2-minus-Case 1 margin gap against the carbon-price axis. The absorption chiller delivers a near-constant 862 GWh_e/yr across the sweep; because that output is not discriminating, panel b instead reports the net cooling-electricity saving after VCC backup and top-up, which rises from \$1.9 M/yr at PUE 1.10 to \$13.8 M/yr at PUE 1.30 and \$23.0 M/yr at PUE 1.50. When the electric chiller is efficient (PUE 1.10) that electricity is cheap to supply, and Case 2 costs \$13.0 M/yr more than Case 1.

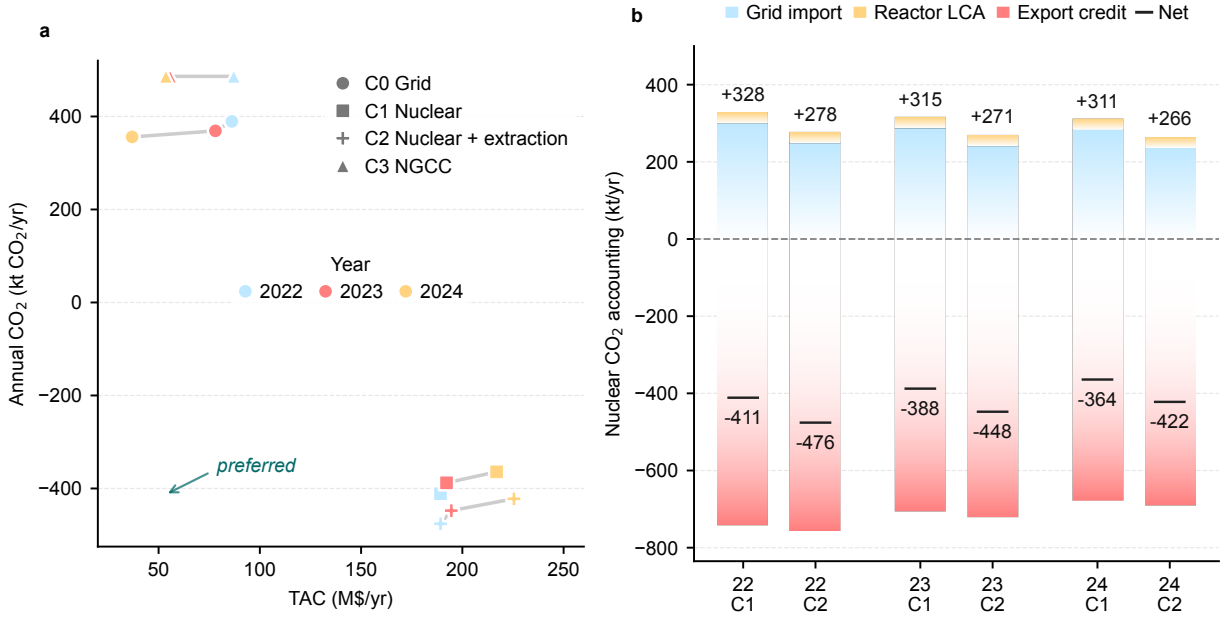


Figure 5: Cost-carbon trade-off and nuclear carbon accounting across ERCOT years 2022, 2023 and 2024. (a) TAC against annual CO₂ emissions for the four configurations. Marker shape denotes case, marker color denotes ERCOT year, and the arrow indicates the Pareto-preferred direction. (b) Carbon-accounting components for the two nuclear configurations: grid-import emissions and reactor lifecycle emissions are offset by grid-export credits; the horizontal mark reports net annual CO₂.

As the chiller becomes less efficient the displaced electricity grows more valuable: the gap narrows to \$4.6 M/yr at PUE 1.30 and reverses to a \$4.6 M/yr saving at PUE 1.50. The main-results baseline uses a more efficient COP-3.2 chiller at a full-load PUE of 1.35, so its \$2.4 M/yr Case 2-minus-Case 1 gap sits between the PUE 1.30 and PUE 1.50 points. Break-even falls near a full-load PUE of 1.40, so absorption is marginally uneconomic at baseline cooling efficiency and pays only when the competing chiller is less efficient. A carbon price narrows the same gap, because Case 2 exports slightly more low-carbon electricity than Case 1, but the effect is modest over the plotted range.

Multi-year electricity price regime.

The ERCOT price regime during the year of operation is the largest single market sensitivity (Figure 8, panel a). The 2022 cluster captures the high natural-gas price regime: delivered natural gas (Henry Hub plus Houston Ship Channel basis) averaged \$6.53/MMBtu, and the NGCC margin falls to -1%, essentially parity with the grid, because the grid clearing price and the on-site fuel cost moved together. In 2023, low gas prices (\$2.63/MMBtu) combined with high ERCOT prices made Case 3 the only on-site option that beat grid supply (a +29% margin). The 2024 cluster shows the deepest negative margin for every on-site option because ERCOT LMPs settled at a multi-year low (~\$28/MWh annual mean), making grid procurement comparatively inexpensive. Across the three years, no on-site option beats grid supply in 2024 or 2022, and only NGCC beats it in 2023. The year regime therefore sets the market boundary for the on-site gas option: a $\pm 50\%$ swing in Henry Hub gas price or ERCOT LMP can overturn the ranking between grid and NGCC supply. Nuclear, by contrast, stays far above grid cost in all three years, so its ranking is governed by the capital-cost rather than the market axis.

Battery storage.

The BESS axis tests whether a 100 MWh / 50 MW lithium-ion battery changes the competitiveness frontier. It lowers TAC by \$0.63 M/yr in each nuclear case, improving the margin by only 0.8 percentage point and never altering the configuration ranking (see Supplementary Information). Because this effect is much smaller than the market, capital-cost and financing axes, it is retained as a secondary boundary rather than as a main panel in Figure 8.

Reactor capital cost trajectory.

SMR overnight capital cost is the dominant technology-cost axis (Figure 8, panel b). At FOAK (\$14,700/kW_e) the margin is −372% (Case 1) and −375% (Case 2): reactor capital outweighs all other costs by roughly a factor of five, and the production tax credit is far too small to matter. At ATB-Mid the margin settles at −146% (Case 1) and −149% (Case 2), so even with the credit, mid-range capital leaves the nuclear cases about 2.5 times more expensive than grid supply. At NOAK (\$2,250/kW_e) the picture inverts: the margin rises to +25% for Case 1 and +22% for Case 2, so both nuclear configurations are cheaper than grid procurement with no carbon price applied. The credit is decisive precisely here, because the pre-credit gap to grid at NOAK capital is small, on the order of 10%, and the \$26.4 M/yr Case 1 credit more than covers it, whereas the same credit barely dents the far larger FOAK and ATB-Mid gaps. Coupling the NOAK reactor with the absorption chiller (Case 2 NOAK) is marginally *less* competitive than Case 1 NOAK, because the absorption capital still slightly outweighs the cooling-side savings, but both stay comfortably below grid cost.

Cost of capital.

The financing assumption changes the same capital-dominated boundary rather than creating a separate operating mechanism (Figure 8, panel c). Sweeping the weighted-average cost of capital between 5% and 10% for Case 2 while holding absorption capital, reactor capital and operating parameters fixed moves TAC from \$168 M/yr to \$251 M/yr, an \$83 M/yr range. The implied grid-cost margin ranges from −115% at 5% WACC to −221% at 10% WACC. By comparison, the full range of group G4 (reactor capital, FOAK to NOAK) moves the Case 2 margin between −375% and +22%, crossing the parity line. Financing materially affects the high-cost FOAK–ATB region but matters less than the reactor capital-cost axis once the trajectory reaches NOAK. Policy or contracting mechanisms that lower the effective cost of capital for SMR projects, such as long-term offtake agreements, sovereign loan guarantees, or accelerated depreciation, can compress but not eliminate the gap to grid supply.

Joint SMR and absorption capital cost frontier.

The joint capital-cost frontier in Figure 8, panel d, turns the preceding one-axis sensitivities into the procurement-relevant capital-cost test. Both axes sweep simultaneously: SMR capital cost runs from NOAK \$2,250/kW_e to FOAK \$14,700/kW_e on the vertical axis, and absorption-chiller capital cost runs from Bare-Low \$450/kW_e to Turnkey-High \$1,200/kW_e on the horizontal axis. Every cell in the 5 × 5 grid is solved at 2023 ATB-Mid market conditions, full-load PUE 1.35, WACC 6.7%, and zero carbon price, with the production tax credit applied. A viable region now opens along the NOAK edge: every cell in the NOAK row reaches a positive grid-cost margin, from +27% with Bare-Low absorption to +14% with Turnkey-High, so the margin = 0 contour enters the published rectangle. The most favorable corner (NOAK reactor with a Bare-Low absorption chiller at \$450/kW_e) lands at +27%, and the least favorable (FOAK reactor with Turnkey-High absorption) at −383%. The frontier is governed almost entirely by the reactor axis: absorption capital shifts the margin by at most 13 percentage points across its full published range, whereas raising reactor capital one trajectory above NOAK drops the margin below −60%. Reaching roughly nth-of-a-kind reactor capital is therefore the binding condition. With the production tax credit it makes colocated SMR with absorption cooling cheaper than grid supply at 2023 ERCOT prices and zero carbon cost, and no tested capital trajectory above the NOAK row is competitive.

Carbon-price crossover.

A carbon price provides a weaker, second route to nuclear competitiveness, distinct from the NOAK capital route just described. Holding capital at the ATB-Mid baseline, the policy axis is the carbon price (Figure 9, panel a). All four cases have net annual CO₂ footprints that are nearly fixed by dispatch (Case 0 +369, Case 3 +486, Case 1 −388 and Case 2 −448 ktCO₂/yr), so each TAC–carbon relationship is essentially linear over the plotted range. The nuclear footprints barely deepen as carbon is valued, because the production tax credit already drives the reactor to export close to its full surplus, leaving little additional export to unlock.

The grid-only Case 0 trends upward at \$0.37 M/yr per \$/tCO₂ because the ERCOT grid mix is carbon-intensive. The two nuclear cases trend *downward*, at about −\$0.39 to −\$0.45 M/yr per \$/tCO₂, because their grid exports displace carbon-intensive generation and receive a corresponding credit. Case 3 NGCC trends upward at \$0.49 M/yr per \$/tCO₂, carrying the full direct emissions on the buyer side. As the carbon price rises, the two on-site options cross the grid-only baseline in opposite directions. At ATB-Mid capital the nuclear cases start well above the baseline and fall below it between \$142 and \$151/tCO₂, while NGCC starts below grid and rises above it near \$191/tCO₂; both crossings are extrapolations beyond the \$0–100/tCO₂ range solved in group G6. Above about \$151/tCO₂ both nuclear cases are cheaper than grid supply at mid-range capital, and by the NGCC–grid crossover near \$191/tCO₂ they are below both grid procurement and the on-site gas alternative. This carbon route is independent of the NOAK capital route: at NOAK capital the nuclear cases already sit below grid cost at zero carbon price.

386 The cogeneration cooling earns only a small, indirect carbon benefit: meeting cooling from steam rather than
387 electricity lets Case 2 both export more and import less than Case 1, giving it a marginally deeper net footprint (−448
388 versus −388 ktCO₂/yr) and a slightly earlier crossover, but it does not create a separate credit mechanism. Both crossings
389 depend on crediting the reactor's merchant exports at the ERCOT hourly average carbon intensity: the nuclear cases
390 reach a net-negative footprint only because the ~1.43 TWh/yr of exported electricity displaces grid generation. The
391 carbon route is correspondingly sensitive to this accounting choice. Halving the export credit roughly doubles the
392 crossover price, and removing it entirely pushes the crossover far above any carbon price modeled here, leaving NOAK
393 reactor capital with the production tax credit as the route that actually makes the nuclear cases competitive.

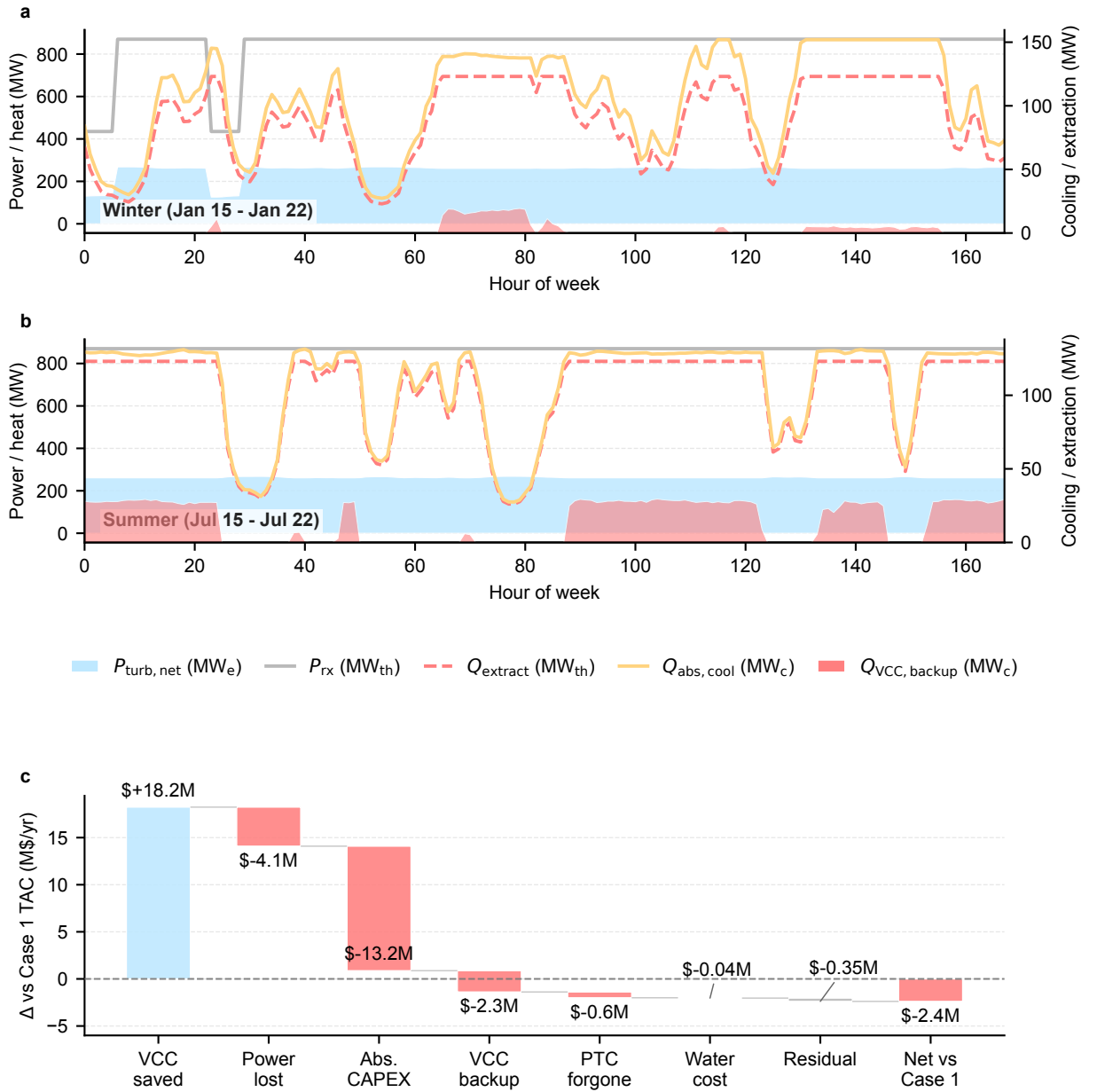


Figure 6: Cogeneration operation and the marginal value of absorption cooling for Case 2. (a) Hourly dispatch over a winter week (15–22 January) and (b) a summer week (15–22 July): reactor thermal output P_{rx} and turbine net electric output P_{tn} are read on the left axis; mid-pressure extraction Q_{ext} , absorption cooling Q_{a} , and VCC backup cooling Q_{v} on the right axis. In both plotted weeks the wet-bulb stays below the 27 °C crystallization gate, so the absorption chiller carries most of the cooling load; the VCC backup tops up on the warmest summer afternoons, when the derated absorber cannot meet demand from steam, and takes over entirely during the roughly 115 crystallization-gate hours per year elsewhere in the record. (c) Waterfall decomposition of the marginal annual value of adding the absorption chiller to a nuclear-only plant (Case 2 against Case 1): positive bars reduce Case 2 TAC, negative bars increase it, “VCC backup” covers steam-limited top-up and crystallization-gate fallback, “PTC forgone” is the production tax credit lost as steam diverts from the turbine, and the residual bar reconciles the component sum with the measured \$2.4 M/yr difference.

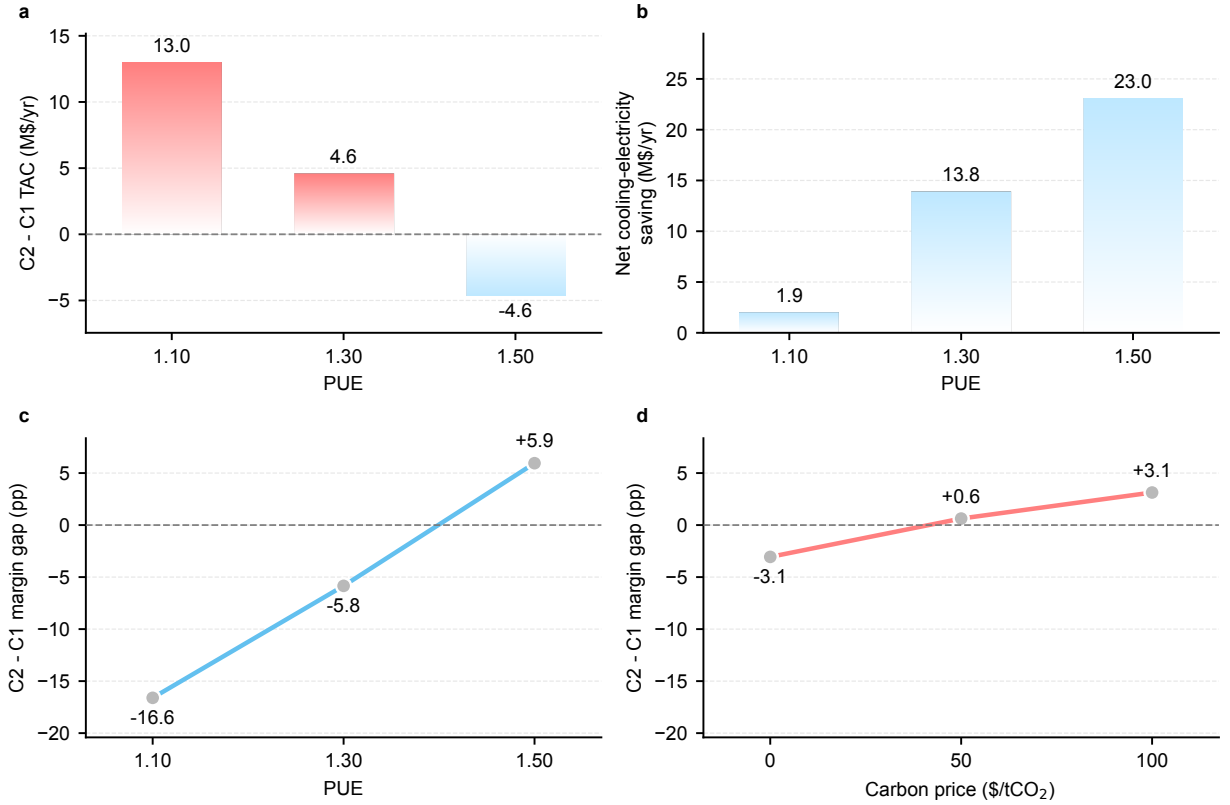


Figure 7: Cooling-side response of the absorption configuration at the 2023 ATB-Mid baseline. (a) Case 2 minus Case 1 TAC difference across full-load PUE; positive is a cogeneration penalty, negative a saving. (b) Net cooling-electricity saving from absorption, measured as gross avoided VCC electricity cost less Case 2 VCC backup/top-up, before fixed absorption capital and turbine-opportunity costs. (c) The same difference expressed as a grid-cost-margin gap across full-load PUE. (d) The margin gap across carbon price. Absorption becomes economic only when the competing electric chiller is inefficient (high PUE) or carbon is priced.

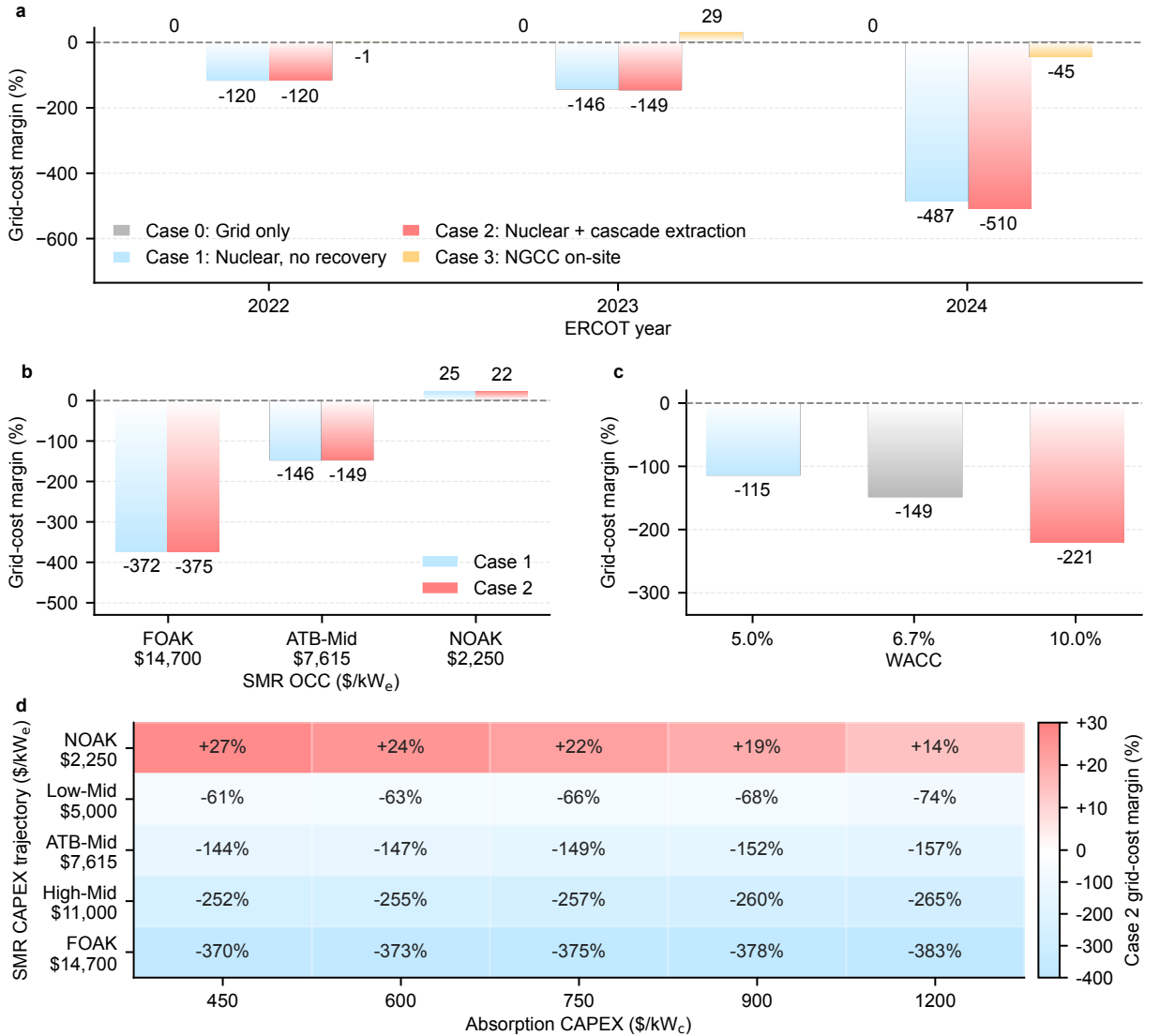


Figure 8: Market, reactor-capital, financing and joint capital-cost boundary atlas. (a) Grid-cost margin for the four configurations across ERCOT years 2022, 2023 and 2024. (b) Grid-cost margin for Cases 1 and 2 across the NREL ATB 2024 BWRX-300 capital-cost trajectories. (c) Case 2 grid-cost margin across the WACC sweep. (d) Joint SMR and absorption-chiller capital-cost viability frontier for Case 2; cell values are grid-cost margin against the 2023 grid-only baseline. Bars or cells below zero indicate configurations that are more expensive than the grid-only baseline.

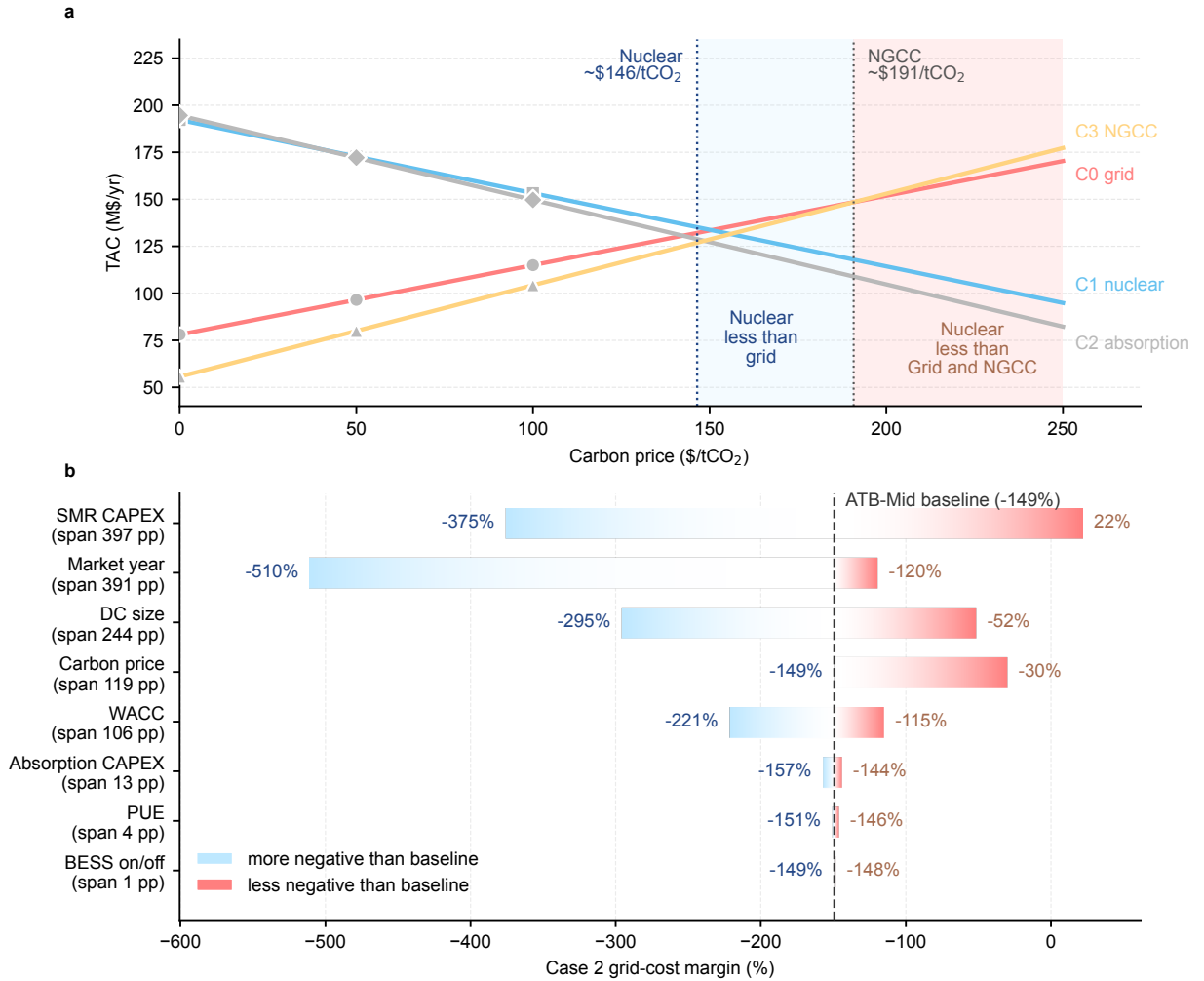


Figure 9: Policy crossover and global sensitivity ranking. (a) TAC extrapolated against carbon price for the four configurations. Vertical dotted lines mark the nuclear and NGCC crossovers against Case 0; the blue region marks where the nuclear cases begin to beat grid supply, and the pink region marks the higher-carbon-price range where they are below both grid and NGCC. (b) Ranked sensitivity of the Case 2 grid-cost margin to each boundary axis, anchored at the 2023 ATB-Mid baseline (-149%).

3.3. Boundary synthesis and limitations

Across the baseline run and eight sensitivity axes, four boundaries define whether and when colocated SMR with absorption cooling reaches grid-supply competitiveness for a hyperscale data center in ERCOT.

The first is *unit-size matching*. At the baseline 94.7 MW_e mean IT load, Case 2 exports about 1.43 TWh/yr net and carries a −149% margin. Scaling the same load shape to 2.5× brings the site close to import-export balance but leaves Case 2 at −62%; scaling to 3.0× improves the margin to −52%. In annual dollars the grid-cost gap stays near \$117–121 M/yr over the 1.0–3.0× range; the visible improvement is the decline in excess cost per MWh of IT load from \$140 to \$49/MWh_{IT}. Size matching therefore removes a major source of distortion but does not by itself open a viable region.

The second is a *market-condition* threshold set by the ERCOT year regime and natural-gas price. At 2024 prices no on-site configuration is competitive; at 2023 prices only Case 3 NGCC is competitive; and at 2022 prices the gas case loses its advantage. This threshold determines the viable region for every on-site option.

The third is a *capital-cost* frontier set by the joint SMR×absorption CAPEX at the baseline load, 2023 market conditions, WACC 6.7%, and zero carbon price, with the production tax credit applied. Here the frontier crosses zero: the entire NOAK reactor row is competitive, from +27% with Bare-Low absorption to +14% with Turnkey-High, so the viable region is the NOAK edge of the published rectangle. The frontier is set almost entirely by the reactor axis, with absorption capital moving the margin by at most 13 percentage points. Roughly nth-of-a-kind reactor capital is therefore the binding condition. With the credit it brings the cogeneration plant below grid cost at 2023 prices and baseline financing with no carbon price, and no tested trajectory above the NOAK row is competitive.

The fourth is a *policy* threshold set by the carbon price under ATB-Mid capital, a weaker second route to competitiveness distinct from the NOAK capital route. At mid-range capital, where the nuclear cases sit far above grid, both cross the grid-only baseline once carbon is priced near \$150/tCO₂, extrapolated beyond the simulated range. The cogeneration cooling earns only a small, indirect carbon benefit, so the policy axis selects the nuclear cases collectively, not Case 2 specifically.

A central result is that the cooling integration in Case 2 is close to cost-neutral and does not, by itself, shift the competitiveness boundary, which is set by reactor capital. In the baseline run Case 2 is 3.1 percentage points worse than Case 1, a \$2.4 M/yr penalty, because the absorption capital, the lost turbine output and the smaller production tax credit together slightly outweigh the displaced-electricity credit (Section 3.1). The sign of this small residual is conditional: the chiller becomes cheaper than the vapor-compression alternative once the full-load PUE exceeds about 1.40, or once a carbon price is applied, but it stays marginally uneconomic at the COP-3.2 baseline. Either way the effect is second-order, because the grid-parity line is governed by reactor capital rather than by the cooling system.

Limitations.

Several limitations constrain the conclusions. The size sweep scales a single hyperscale load shape uniformly; sites with markedly different temporal structure, such as mixed AI training and inference clusters, or different climates would shift the absorption value proposition. The carbon-price analysis applies one uniform price to both the average-intensity export credit and the on-site direct-emissions debit, a strong assumption about how a future carbon policy would be implemented. The hourly average emission factor is also a proxy for the true marginal factor; a merit-order marginal series would refine the avoided-emissions credit.

Other omissions act in one direction. Demand charges and ancillary-service revenues are not modeled. Restoring the full delivered grid cost, including transmission, ancillary and demand charges, would raise the Case 0 baseline and narrow every on-site margin, so the penalties reported here are conservative against the on-site configurations.

A separate caveat concerns the on-site gas comparator. Case 3 captures the part-load heat-rate penalty but omits forced-outage modeling, a redundant unit, a grid tie and outage cost, whereas Cases 1 and 2 retain grid import to cover outages. Meeting the five-nines availability that hyperscale operators target would raise the Case 3 cost and narrow its 29% advantage.

A further caveat concerns the production tax credit. We apply it at full prevailing-wage value to all net nuclear generation under an assumed power-purchase-agreement structure, hold it at nominal 2024 amounts without inflation indexing, and drive it from the hourly market price as a gross-receipts proxy. A behind-the-meter structure that disqualified self-consumed generation, or expiry after 2032, would weaken the NOAK competitiveness reported here. Two final caveats follow from the analysis above: the reported margin reflects the smallest deployable reactor rather than an optimized multi-module buildout, and the carbon-price route depends on crediting merchant exports at the grid-average intensity. All of these should be evaluated with region-specific tariff, market-service and marginal-emissions data before procurement.

4. Conclusion

Across 100 annual hourly scenario runs, colocated SMR with double-effect absorption cooling is a conditional rather than a categorical cost-competitive solution for an ERCOT data center. The runs span the four-case baseline, data-center size, three ERCOT years, three SMR capital-cost trajectories, a 5×5 joint reactor-absorption capital-cost grid, three carbon prices and three WACC levels, with the Section 45U production tax credit applied to the nuclear cases. At 2023 ATB-Mid prices for the baseline 200 MW_e campus, the cogeneration configuration is 149% more expensive than grid supply even with the credit, and about \$2.4 M/yr more expensive than an SMR-only plant; the absorption chiller is close to cost-neutral, its capital roughly offsetting the cooling electricity it displaces. The decisive lever is reactor capital combined with the production tax credit. At first-of-a-kind and mid-range capital the SMR cases remain far above grid. At nth-of-a-kind capital the production tax credit brings both below grid cost, by 25% (Case 1) and 22% (Case 2) with no carbon price; the entire NOAK row of the joint capital-cost grid is competitive, so the viable region opens at its nth-of-a-kind edge. A carbon price near \$150/tCO₂ is a weaker second route at mid-range capital, but it is extrapolated beyond the simulated range and depends on crediting merchant exports at the grid-average carbon intensity. The cooling integration earns only a small, indirect carbon benefit over the SMR-only option, not a separable credit.

Competitiveness therefore turns on reactor capital reaching the nth-of-a-kind trajectory in combination with the production tax credit. Under those conditions, a load sufficient to absorb the smallest commercial reactor and a favorable ERCOT price regime are enough to bring the cogeneration plant below grid cost without any carbon price. At first-of-a-kind or mid-range capital, none of these conditions, and no plausible carbon price, makes the plant competitive. Absorption cooling is close to cost-neutral throughout and does not move the boundary, which is set by reactor capital. The scenario envelope and input provenance documented in the Supplementary Information provide the basis for extending the boundary analysis to other ISO regions, climates, and data-center sizes.

CRedit authorship contribution statement

Honglin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Buxin She:** Methodology, Validation, Writing – review & editing. **Jie Zhang:** Conceptualization, Supervision, Resources, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All input parameters, scenario definitions, run-level summaries, hourly dispatch traces, and figure source data that support the findings of this study are openly available in a persistent public archive (Zenodo; DOI to be assigned at acceptance) and are described in the Supplementary Information. ERCOT day-ahead locational marginal prices are sourced from gridstatus.io under their public API terms; hourly average grid carbon intensities are derived from EIA-930 generation mix and UNECE 2022 lifecycle emission factors; Houston wet-bulb temperature is sourced from the Open-Meteo historical-weather endpoint; reactor and NGCC capital costs are sourced from the NREL Annual Technology Baseline 2024 at <https://atb.nrel.gov/electricity/2024>; absorption-chiller capital costs are bracketed using published list-price ranges from the ASHRAE TC8.3 absorption-chiller technical-data guide and publicly available vendor literature. No proprietary or restricted-access data are used in this study.

Acknowledgements

The authors thank the GE–Hitachi BWRX-300 design-basis publications and the NREL Annual Technology Baseline team for the open reactor capital cost trajectories that anchor this analysis, and the Open-Meteo and gridstatus.io contributors for the open weather and electricity-market data feeds that made the multi-year sensitivity analysis tractable. Gurobi Optimization is acknowledged for the academic license that enabled the linear programming runs.

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